

Average Specific Angular Momentum of Milky Way and Andromeda Merged Dark Matter Halo Analysis

LAINÉ KOWALSKI¹

¹*University of Arizona*

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1. INTRODUCTION

A galaxy’s dark matter halo is an invisible spherical cloud of dark matter that surrounds a galaxy, extending far beyond the visible stars, and is crucial for stabilizing the galaxy’s structure. Dark matter halos drive galaxy evolution by dominating the gravitational forces that govern a galaxy’s orbital dynamics. Halos play a critical role in shaping the outcome of galactic mergers, such as for merging dark matter halo components of the Milky Way (MW) and Andromeda (M31) galaxies, which will eventually result in a single, uniform halo.

In terms of galaxy evolution, the process by which galaxies change and develop over time due star formation, interactions with other galaxies, and accumulation of gas and dark matter, determining the post-merger physical attributes of dark matter halo particles within the combined MW-M31 system is critical for understanding how galaxies merge and behave dynamically. Examining specifically how the dark matter halo components interact during a merge interaction may also help to shed light on the nature of dark matter and its central role within galaxy formation, evolution, and internal galactic interactions. Studying how dark matter halos interact through the merging process helps

astronomers piece together the evolution and future properties of these two galaxies as they become one, which results in a better understanding of galaxy merger dynamics outside of our galactic Local Group as well. Further understanding of dark matter also has broader implications for particle, nuclear, and other areas of contemporary physics.

Regarding dark matter halo dynamics and its role in galaxy evolution, the literature states that dark matter halos are essential galactic structures that surround/host galaxies (J. Dubinski 1992). Galaxies form at the center of dark matter halos, and dark matter halo development/evolution is consistent in galaxies across the universe (C. Frenk & S. White 2012). Therefore, dark matter plays a significant role in shaping all galaxies, regardless of their size or brightness (J. Wolf et al. 2010). Initial central galaxy formation affects the shape and mass of the surrounding dark matter halo because as baryons (normal matter) gather at the center of a halo, the shape of the halo becomes more symmetric and less elongated (M. G. Abadi et al. 2010). Dark matter halos form and grow, starting as small structures that stop expanding, collapse and then merge or gather more dark matter to become larger, in a process known as hierarchical growth (C. Frenk & S. White 2012). Merging halos of similar size, such as

MW and M31, will thoroughly mix their layers (C. Frenk & S. White 2012). Furthermore, the presence/quantity of dark matter in spiral galaxies observed using the Tully-Fisher relation, which relates halos and the galaxy's rotational velocity, proved that faster rotating galaxies have more massive dark matter halos as a galaxy's outer regions are most affected by the presence of dark matter (C. Frenk & S. White 2012). In terms of halo mergers, energy involved in the merger and the way halos spin around each other greatly influence the final shape and size of a merged halo. Final merged halos can be elongated or flattened, and the size increases in a predictable way based on the energy and spin from before the merger (N. E. Drakos et al. 2019). Dark matter also has significant implications for a galaxy's angular momentum. The general trend in the universe is that angular momentum is transferred from central regions to outer edges during galactic mergers (S. Peirani et al. 2004). Therefore, a significant increase in angular momentum at greater distances, especially after a merger, is expected. Therefore, average angular momentum should peak at further radii for merged galaxies than compared to the pre-merger galaxies. Dark matter halos are categorized into two types: "accretion" halos, which grow smoothly, and "merger" halos, which experience immediate growth by quickly absorbing large amounts of mass. Over time, all halos increase their spin, but halos formed through mergers gain more spin relative to their size than those formed through steady accretion (S. Peirani et al. 2004). Another study found that cold flow gas, gas accreted along cosmic filaments, enters a galaxy with much higher specific angular momentum compared to dark matter (K. R. Stewart et al. 2013). This shows that dark matter components are more uniform than varied dynamic gaseous components (K. R. Stewart et al. 2013). Finally, the shape and rotation of cosmic halos are influenced by the

direction and strength of tidal forces, which primarily cause halos to spin smoothly along specific axes, helping to explain galaxy rotations (J. Dubinski 1992). Overall, dark matter halos are the static framework for galaxy creation and are the ultimate determining factor for galaxy's physical and dynamical attributes, predominantly angular momentum and spin.

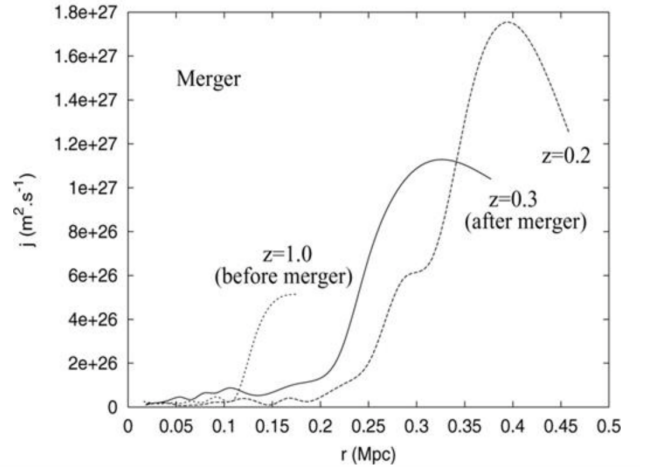


Figure 1. Figure 6 from Ref. () This graph demonstrates how the average angular momentum of the dark matter halo changes as a function of the radius before and after a merger. The first curve, before merger, represents an arbitrary galaxy's angular momentum as peaking earlier compared to the post merger curve, which peaks at a further radii. This indicates that angular momentum is transferred to a merged galaxy's periphery and generally increases in magnitude once merged, as indicated by the higher peak.

Several open questions still exist regarding dark matter halo mergers, which are crucial for advancing our understanding of galaxy formation and structure. Key inquiries focus on how these mergers affect the halo mass function over time, the specific properties of dark matter that influence merger dynamics, and the detailed physical processes involved, including the transfer of angular momentum (J. S. Bullock & M. Boylan-Kolchin 2017). Questions remain about the effects of baryonic matter and feed-

back mechanisms from star formation and black hole activity on these mergers (J. S. Bullock & M. Boylan-Kolchin 2017). In addition, understanding the time it takes for the merged dark matter halos to reach equilibrium is a significant area of ongoing study. These questions highlight the complexity of halo mergers and their fundamental role in cosmological research.

2. PROJECT

This project will investigate the dynamics of the merged dark matter halo versus what the MW halo and M31 halo dynamics were individually before the merger. I will compute the average angular momentum of MW, M31 and the merged Milkdromeda dark matter halo at different radii graphically to observe where average angular momentum peaks for each. I will observe how the average angular momentum differs before the merger, when MW and M31 are individual galaxies, and when the galaxies combine to form Milkdromeda. These three results will be compared to explore if the profiles are similar/different and the reason why.

Overall, this project addresses questions about the physical processes involved in a dark matter halo merger, and how the MW-M31 merger affects average angular momentum of the halo particles, as mentioned in Paragraph 4 of the Introduction. Specifically, I am investigating if the average angular momentum magnitude as a function of galactic radius increases post-merger? And also, is the peak average angular momentum at the same or different radius compared to the peak average angular momentum of the dark matter halos in either galaxy before they merged and why?

Answering questions about the physical processes involved in a dark matter halo merger, specifically regarding how the MW-M31 merger affects the average angular momentum of the halo particles, is critical for several reasons in the context of galaxy evolution. Studying changes in angular momentum due to galaxy

mergers helps us to understand the dynamic processes that drive galaxy evolution. Angular momentum is a fundamental property that affects the shape and structure of galaxies, influencing everything from star formation to the distribution of matter. Investigating how the average angular momentum changes across galactic radii post-merger could help infer how mergers reshape galaxies, such as an increase in angular momentum at larger radii might suggest more extended/disrupted galactic disks, impacting the overall structure of the merged galaxy. The location of the peak average angular momentum, depending on if it shifts or remains the same post-merger, provides insights into the effectiveness of theoretical models in predicting the outcomes of major merger events. Also, since dark matter constitutes a significant portion of a galaxy's mass and influences its gravitational field, understanding its role during mergers could reveal more about the nature of dark matter.

3. METHODOLOGY

This project utilizes simulation details from literature by Besla +2012 to investigate changes in average angular momentum pre and post merger (G. Besla et al. 2012). The N-body simulation data in this paper models the gravitational interactions of multiple bodies, such as multiple galaxies and different particle types within each galaxy, to understand their dynamics and evolution over time. Before these two galaxies are merged, initially, the Milky Way is a barred spiral galaxy with a central bulge and a supermassive black hole at the center, a prominent stellar disk with spiral arms with a diverse range of stars/matter, and a surrounding, elongated dark matter halo. Andromeda is similar in size to the Milky Way, as it is also a barred spiral galaxy with a central bulge, surrounded by a large stellar disk with spiral arms rich in young stars, gas, and dust. It also has an extensive dark matter halo, and ongoing research

is still determining its exact halo shape. Both galaxies' dark matter halos are quite extensive and are most likely slightly elongated. These two galaxies will merge in around 6.7 Gyrs, and are part of our galactic Local Group, along with other satellite and dwarf galaxies.

The project code will analyze the dynamics of the merged dark matter halo compared to the pre-merger dynamics of the Milky Way (MW) and Andromeda (M31) halos. It selects two points in time, before the merger (Snapshot 0) and post-merger (Snapshot 680), calculates the average angular momentum magnitude at various radii for MW, M31, and the merged Milk-dromeda, and compares these profiles to understand their differences. From the simulation data, I will only utilize particle type 1, which are dark matter halo particles of all equal mass. Only the LowRes simulation data is needed for this project.

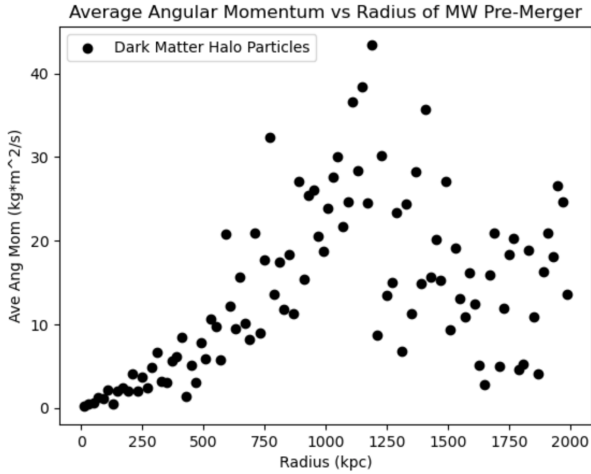


Figure 2. Figure 2 () My graph models the average angular momentum of the dark matter halo as a function of the radius before the merger for the MW galaxy. The peak average angular momentum at a specific radius (peak of the curve of best fit) will be plotted and identified. This graph will be compared to M31's and Milk-dromeda's average angular momentum profiles to determine how average angular momentum changes within the merged galaxy

The functions in my code calculate the center of mass position r_{COM} using the equation for center of mass, where R_i is the position vector of particle i , and N is the total number of particles within a maximum radius, R_{max} . To improve accuracy, the code iteratively reduces R_{max} and recalculates r_{COM} until the value converges, or stops changing. This method ensures a stable center of mass for both pre and post merger galaxies, which is necessary for calculating angular momentum relative. To compute the absolute average angular momentum L_{ave} at a given radius r , the code calculates the cross product of position and velocity vectors for particles within a radial bin, relative to the system's center of mass. Each term $(\mathbf{r}_i - \mathbf{r}_{COM}) \times (\mathbf{v}_i - \mathbf{v}_{COM})$ gives the specific angular momentum of particle i , which is averaged over all N_{star} particles in the bin and multiplied by the mass, m , of particle type 1. This equation in the code computes average angular momentum at different radii, helping to reveal the rotational structure/dynamical evolution of the dark matter halo.

Three different plots will be generated through the code demonstrating dark matter halo average angular momentum at various radii for MW and M31 individually pre-merger and for the merged Milk-dromeda halo. These plots are scatter plots, where data points represent the magnitude of average angular momentum at a specific radius. These scatter plots will include curves of best fit to determine at what radii the peak angular momentum is and if it is a larger magnitude post-merger. Code from Homework6's OrbitCom notebook is also imported to determine where the post-merger snapshot is graphically (at 6.7 Gyrs) and the Homework5 Mass Profiles for Milky Way components graph was utilized to determine the radius at which around 80% of the dark matter halo mass is enclosed (around 200 kpc). This

graph helped to determine the virial radius set in my code, setting a limit on how far r should extend.

The literature states that the transfer of angular momentum, especially after major mergers, moves from the central regions to the periphery and is larger in magnitude (S. Peirani et al.

2004). Therefore, after the dark matter halos of MW and M31 merge into one, I hypothesize that the peak average angular momentum should occur at a further radius from the center and will have a greater magnitude than compared to the two pre-merger, individual galaxies.

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